

MLE for linear and logistic regression

Choose a distribution for y given x . Its negative log-likelihood is the loss; minimising it fits the model.

THE QUESTION

Where do squared error and cross-entropy come from, and what replaces them when the noise is not a plain Gaussian?

THE INTUITION

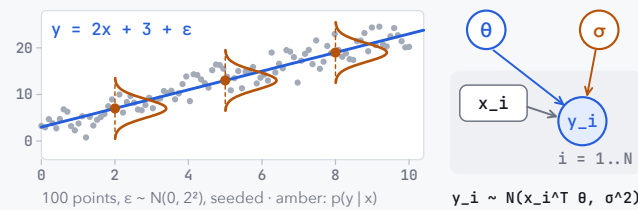
Hold x fixed and treat y as a draw from a distribution centred on the model's prediction. Multiply the densities over the points and take $-\log$: Gaussian noise gives squared error, Bernoulli labels give cross-entropy.

KEY TAKEAWAY

Choosing a loss is choosing a noise model. Gaussian: least squares, then $\hat{\sigma}^2 = \text{mean squared residual}$. Laplace: absolute error. Bernoulli: cross-entropy.

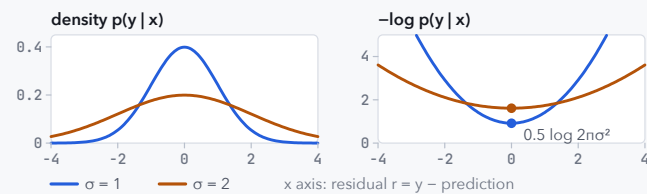
noise model $p(y | x, \theta)$ → product over points → $-\log$: a sum → the loss → minimise: closed form or iterate → $\hat{\theta}$, then $\hat{\sigma}^2$

1 Linear regression puts a Gaussian on each y



The model: $y = x^T \theta + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, so $p(y | x, \theta) = \mathcal{N}(y | x^T \theta, \sigma^2)$. Inputs are held fixed; only y varies. Given θ the points are independent: $p(y | X, \theta) = \prod_i p(y_i | x_i, \theta)$. These 100 points give $\hat{\theta} = (3.02, 2.03)$ and $\hat{\sigma} = 1.92$ (true 3, 2 and 2).

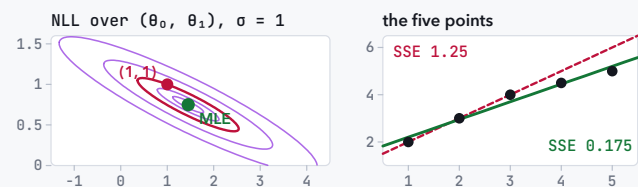
2 $-\log$ turns the product into squares



$$\text{NLL}(\theta) = \sum_i \frac{(y_i - x_i^T \theta)^2}{2\sigma^2} + \frac{N}{2} \log(2\pi\sigma^2)$$

The log is increasing, so the most likely θ minimises the NLL. σ scales the squares and adds a constant, so $\hat{\theta}$ is least squares for every σ . Check (quoted, log-likelihood notebook): noise sd 0.1, $\sigma = 1$ in the NLL, true line: $92.39 = 50 \log 2\pi + 0.5 \cdot 100 \cdot 0.1^2$.

3 Five points: the closed form by hand

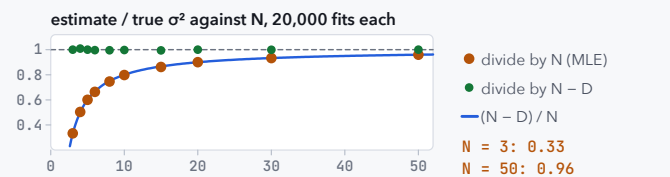


The slider notebook's $x = 1, \dots, 5$, $y = (2, 3, 4, 4.5, 5)$, plus a column of ones:

$$\hat{\theta} = (X^T X)^{-1} X^T y = \frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix} \begin{pmatrix} 18.5 \\ 63 \end{pmatrix} = \begin{pmatrix} 1.45 \\ 0.75 \end{pmatrix}.$$

A quadratic NLL has elliptical contours; the long axis trades intercept against slope. The notebook's "true values" (1, 1) were never used to make the data; they sit 0.54 nats above the minimum.

4 The MLE of σ^2 is the mean squared residual

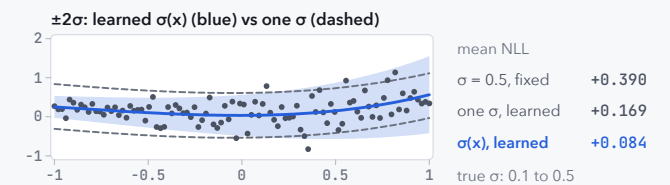


Set $\partial \text{NLL} / \partial \sigma^2 = -\text{SSE} / (2\sigma^4) + N / (2\sigma^2) = 0$ at $\hat{\theta}$:

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{N} \sum_{i=1}^N (y_i - x_i^T \hat{\theta})^2 = \frac{\text{SSE}}{N}.$$

Five points: $0.175/5 = 0.035$. The fit absorbs part of the noise: $\mathbb{E}[\hat{\sigma}^2] = (N - D)\sigma^2 / N$ for D weights. Divide by $N - D$ (0.058) instead, like $n - 1$ on the MLE sheet.

5 Heteroscedastic noise: learn $\sigma(x)$ too

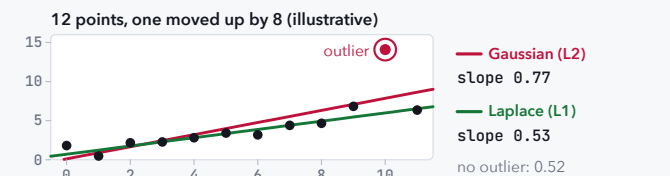


Give each point its own variance and fit both parts by gradient descent:

$$\text{NLL} = \sum_i \left[\frac{(y_i - \mu(x_i))^2}{2\sigma(x_i)^2} + \log \sigma(x_i) \right] + \frac{N}{2} \log 2\pi.$$

Noisy points count less; $\log \sigma$ stops σ growing without limit. Output $\log \sigma$ so $\sigma > 0$. Here μ is cubic, $\log \sigma = a + bx$ (Adam); $\hat{\sigma}$ runs 0.14 to 0.49. The notebook's network, quoted: 0.33, -0.25 , -0.49 .

6 Laplace noise gives absolute error

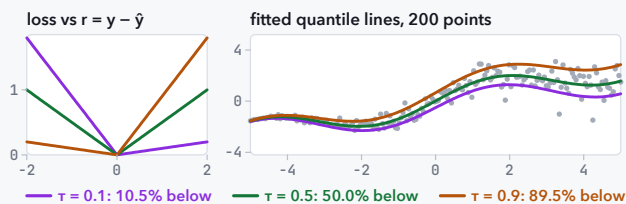


$p(y | x) \propto e^{-|y - x^T \theta| / b}$ gives $\text{NLL} = \frac{1}{b} \sum_i |y_i - x_i^T \theta| + N \log 2b$. A squared penalty pulls a point in proportion to its residual; the absolute one caps the pull at ± 1 . For a constant model the Gaussian MLE is the mean, the Laplace MLE the median. No closed form: IRLS or a linear program.

Other likelihoods, other losses

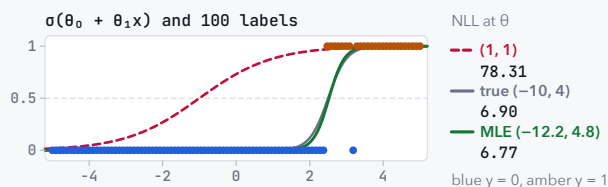
An asymmetric loss fits a quantile, a Bernoulli likelihood gives cross-entropy, separable labels break the MLE.

7 The pinball loss fits a chosen quantile



$\rho_\tau(r) = \max(\tau r, (\tau - 1)r)$, $r = y - \hat{y}$, costs τ per unit of under-prediction and $1 - \tau$ per unit over, so the best line leaves a fraction τ below it; $\tau = 0.5$ gives the median. It is an asymmetric Laplace NLL. The quantile notebook's data (noise sd $0.1(x + 5)$), features 1, x , $\sin x$; its network (quoted) leaves 8%, 49%, 90% below.

8 Labels 0/1: one biased coin per input

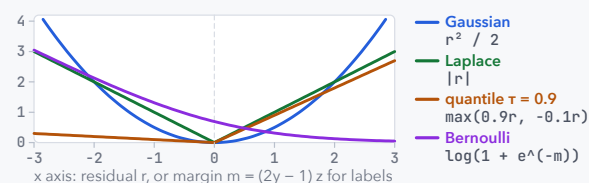


A coin gives $L = \prod_i \theta^{y_i} (1 - \theta)^{1-y_i}$. Logistic regression gives each input its own coin, $\hat{y}_i = \sigma(x_i^T \theta)$:

$$\text{NLL}(\theta) = - \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)],$$

the binary cross-entropy (times N). Our seeded labels differ from the log-likelihood notebook's, yet both NLLs match its printout (78.31, 6.90): both sets have 25 ones and the same $\sum_i y_i x_i$. The likelihood sees the labels only through $X^T y$.

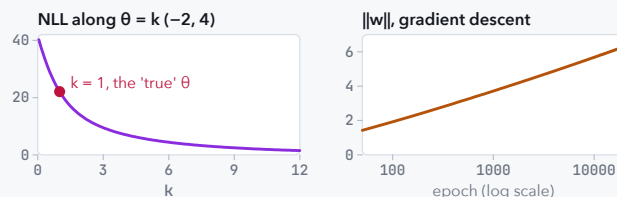
9 Same recipe, four penalties on the residual



noise	$-\log p$, one point	gradient in θ
Gaussian	$r^2 / (2\sigma^2)$	$-rx / \sigma^2$
Laplace	$ r / b$	$-\text{sign}(r)x / b$
pinball	$\rho_\tau(r) / b$	$-(\tau - \mathbb{1}[r < 0])x / b$
Bernoulli	$\log(1 + e^z) - yz$	$(\sigma(z) - y)x$

With $r = y - x^T \theta$ and $z = x^T \theta$, constants dropped. Only the Gaussian gives equations linear in θ , hence a closed form; the rest need gradient descent, Newton's method or IRLS.

10 Separable labels have no finite MLE



If a line splits the classes, scaling θ up pushes every $\hat{y}_i$ towards its label, so the NLL falls towards 0 and never gets there. The slider notebook's 60 points ($y = 1$ exactly when $x > 0.5$) score 22.1 at its "true values" $(-2, 4)$ and 2.07 at $(-20, 40)$. The two-cloud notebook's loss keeps falling (quoted: 0.142 at epoch 50, 0.012 at 950) as the weights grow like $\log t$. Stop early or add a prior (the MAP estimation sheet).

11 The NLLs in NumPy

```
import numpy as np

def nll_gauss(th, s2, X, y):
    r = y - X @ th
    return r @ r / (2 * s2) \
        + len(y) / 2 * np.log(2 * np.pi * s2)

def nll_bern(th, X, y):
    z = X @ th # logits
    # log(1 + e^z) - y z, never overflows
    return (np.logaddexp(0, z) - y * z).sum()

th = np.linalg.solve(X.T @ X, X.T @ y)
s2 = np.mean((y - X @ th)**2) # MLE
```

Solve the normal equations rather than invert $X^T X$; the linear regression notebook adds 10^{-8} to the diagonal before a Cholesky solve. Working from logits avoids $\log 0$ when $\sigma(z)$ rounds to 1.

12 Pause and think

Does σ^2 move the least-squares line? No: it scales the NLL and adds a constant.

Why does $\hat{\sigma}_{\text{MLE}}^2$ run low? The fitted line absorbs part of the noise. Divide by $N - D$.

A sensor sometimes returns garbage: Gaussian or Laplace? Laplace, since one point's pull stays bounded.

Pinball loss with $\tau = 0.5$? The conditional median, the Laplace fit.

Two label sets share $X^T y$: same logistic MLE? Yes, the likelihood is identical.

Separable data: where is the minimum? Nowhere finite; the NLL tends to 0 as $\|\theta\| \rightarrow \infty$.

noise model $\rightarrow$ $-\log$ likelihood $\rightarrow$ loss • Gaussian: least squares, $\hat{\sigma}^2 = \text{SSE} / N$ • Laplace: absolute error • pinball: quantiles • Bernoulli: cross-entropy

Source: lecture slides "Maximum likelihood estimation for linear and logistic regression" and its notebooks (normal likelihood, log-likelihood surfaces, regression sliders, logistic MLE, quantile regression, noise models for neural networks), nipunbatra.github.io/pml-teaching. Unquoted numbers: NumPy with fixed seeds. The σ^2 estimate, Laplace noise and the gradient table go beyond the slides. Corrections: the NLL equals $\|y - X\theta\|^2 / (2\sigma^2)$ plus a constant (not "proportional to" it), and the cross-entropy box needs brackets round both terms.